HOL2

AI in GitHub Actions

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Git

Git은 형상 관리 도구 중 하나로, 소스 코드의 변경 이력을 로컬과 분산 환경에서 관리할 수 있게 해주는 **분산 버전 관리 시스템**이다.

- 코드의 변경 이력 관리
- 오픈소스 기반
- 분산 개발 지원: 중앙 서버 없이도 로컬에서 전체 이력을 보유해 오프라인 개발이 가능
- Commit / Branch / Merge를 통한 버전 관리

GitHub

깃허브(GitHub)는 분산 버전 관리 툴인 깃(Git)을 호스팅하는 협업 플랫폼

- Git 저장소를 중앙에서 관리하여 개발자가 함께 코드를 작성하고 공유할 수 있음
- Pull Request를 통해 변경 사항을 리뷰하고 코드 품질을 유지, Issues로 버그/기능 개발을 관리
- CI/CD – GitHub Actions를 통해 빌드, 테스트, 배포를 자동화
- AI 기발 개발 (GitHub Copilot) – 코드 리뷰 / 보안

GitHub Basic 용어

- Repository (저장소) – 파일이나 폴더를 저장해두는 저장소
- Clone – GitHub의 원본 저장소를 로컬에 복제
- Fork – 다른 사람의 Repository를 내 Repository로 복제
- Add – 작업 디렉토리 영역의 변경 내용을 스테이징에 넘김
- Commit – 파일을 추가, 변경한 내용을 저장소에 저장하는 작업
- Push – 추가, 변경한 내용(Commit 한 내용)을 원격 저장소에 업로드하는 작업

- Branch – Feature, Bug Fix등 독립적인 개발 라인
- Pull Request (PR) – GitHub 브랜치에 Push한 변경 사항을 다른사람에게 알림

```
GitHub Repository (Remote)
  ↓
Clone
  ↓
Local Workspace
  ↓
Create Branch
  ↓
Edit Files
  ↓
Add
  ↓
Commit
  ↓
Push (to branch)
  ↓
Pull Request (PR)
  ↓
Merge to Main
```

Hello GitHub Actions

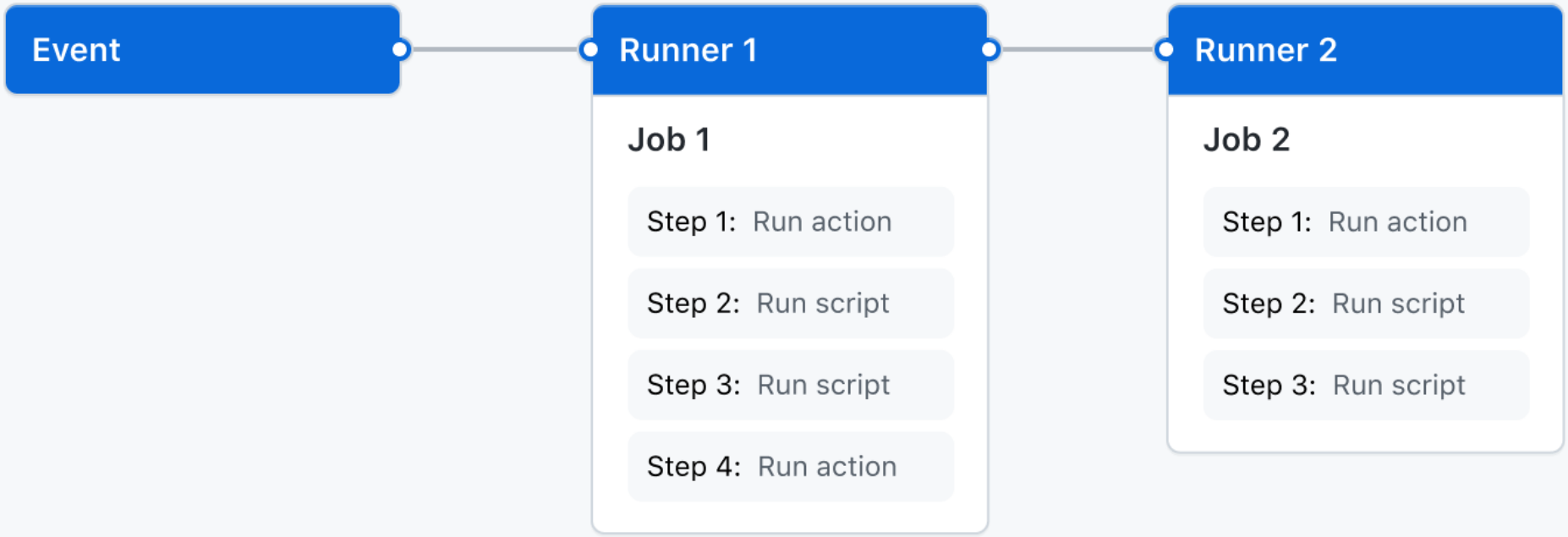
- **GitHub에 기본 Built-in된 CI/CD**
  - ✓ **GitHub Enterprise Cloud** 고객을 위한 호스팅
  - ✓ **GitHub Enterprise Server**를 통한 자체 호스팅 가능

- **YAML로 워크플로 작성**
  - ✓ 워크플로우는 jobs으로 구성되며, jobs는 steps으로 구성됨
  - ✓ 각 단계는 **재사용 가능한 액션** 또는 커스텀 스크립트를 포함

- **Runner에서 작업 실행**
  - ✓ GitHub가 클라우드에서 관리하는 GitHub-Hosted Runners 사용
  - ✓ 직접 설계, 구현 및 관리하는 Self-Hosted Runners 사용 가능

- **Built-in Security**
  - ✓ 통합된 Secret Store와 OpenID Connect(OIDC)를 사용하여 환경 변수와 자격 증명을 안전하게 관리

GitHub Actions Components

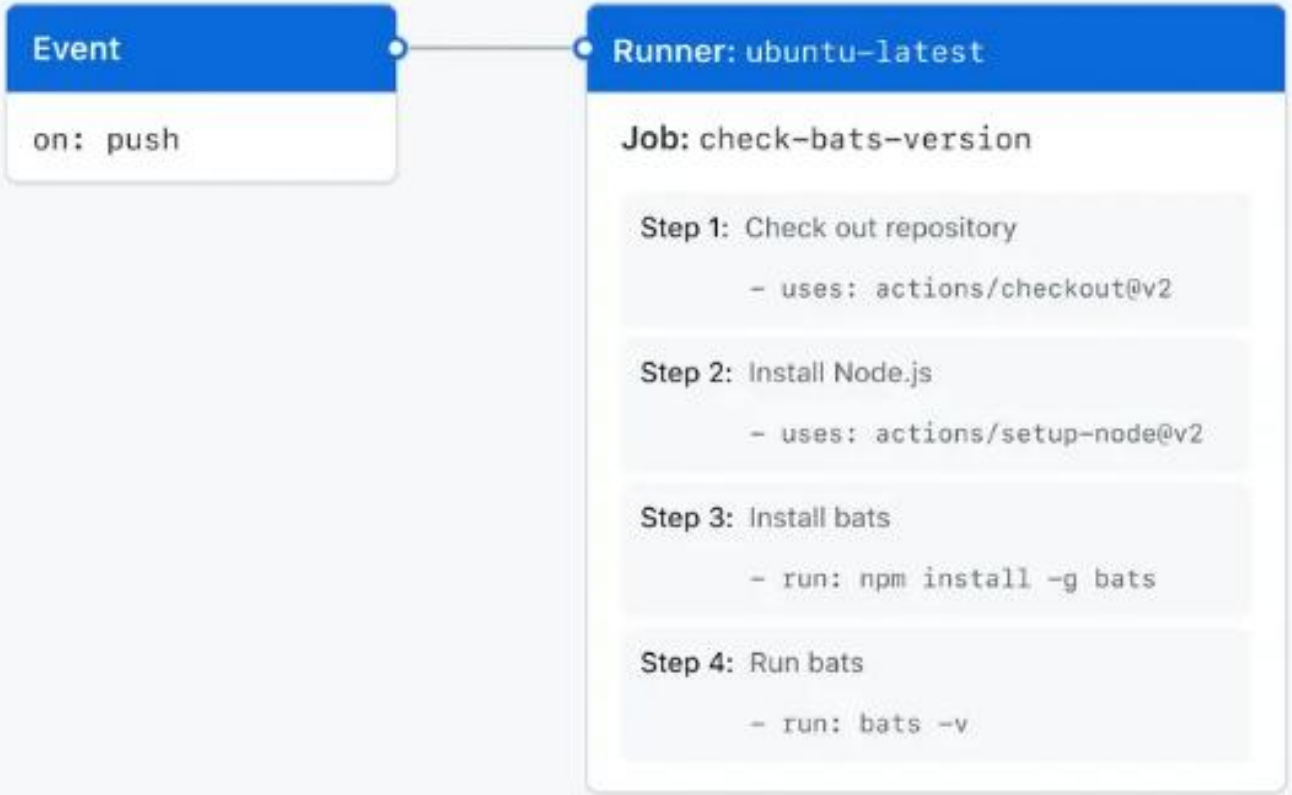


- **Workflow**: .yml 파일에 정의된 최상위 자동화 프로세스입니다.
- **Job**: 동일한 러너에서 실행되는 “Step”들의 셋으로, 작업은 병렬 또는 순차적으로 실행될 수 있습니다.
- **Step**: 명령을 실행할 수 있는 개별 작업이나 "Action"입니다.
- **Runner**: 워크플로우가 실행되는 서버/환경입니다.

GitHub Actions - Workflow

YAML

```yaml
name: learn-github-actions
run-name: ${{ github.actor }} is learning GitHub Actions
on: [push]
jobs:
  check-bats-version:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - uses: actions/setup-node@v3
        with:
          node-version: '14'
      - run: npm install -g bats
      - run: bats -v
```



- ✓ GitHub Actions ***workflow***는 repository에 ***event***가 발생할때 Trigger된다. (예: Pull Request, Push)
- ✓ Workflow는 한개 이상의 ***jobs*** 를 포함한다.
- ✓ 각 Job안에는 한개 이상의 ***steps***이 있고, 각 step은 ***action***을 정의 (define)하거나 실행 (run)하는 스크립트를 포함한다.

Go to:

https://github.com/skills/ai-in-actions

COPY EXECISE 클릭

AI in Actions

Learn how to integrate AI models directly into your GitHub Actions workflows using GitHub Models.

Welcome

- **Who is this for**: Developers new to AI integration who want to learn how to incorporate AI capabilities into their GitHub Actions workflows
- **What you'll learn**: How to use GitHub Models service to add AI inference to your workflows, pass dynamic data to AI prompts, and combine AI actions with other GitHub Actions
- **What you'll build**: A basic AI question-answering workflow and an automated issue analysis system that comments on new issues
- **Prerequisites**: Basic familiarity of GitHub Actions (e.g taught in Hello GitHub Actions exercise)
- **How long**: This exercise takes less than 30 minutes to complete.

In this exercise, you will:

1. Set up a basic AI inference workflow with proper permissions
2. Learn how to combine AI inference with other GitHub Actions to create intelligent automation workflows
3. Build an automated AI issue analysis workflow

How to start this exercise

Simply copy the exercise to your account, then give your favorite Octocat (Mona) **about 20 seconds** to prepare the first lesson, then **refresh the page**.

COPY EXERCISE →

1 **General**

Owner *

vitoc /

Repository name *

skills-ai-in-actions

skills-ai-in-actions is available.

Great repository names are short and memorable. How about glowing-octo-computing-machine?

Description

Exercise: AI in Actions

23 / 350 characters

2 **Configuration**

Choose visibility *

Choose who can see and commit to this repository

Public

3 **Jumpstart your project with Copilot (optional)**

Tell Copilot coding agent what you want to build in this repository. After creation, Copilot will open a pull request with generated files - such as a basic app, starter code, or other features you describe - then request your review when it's ready.

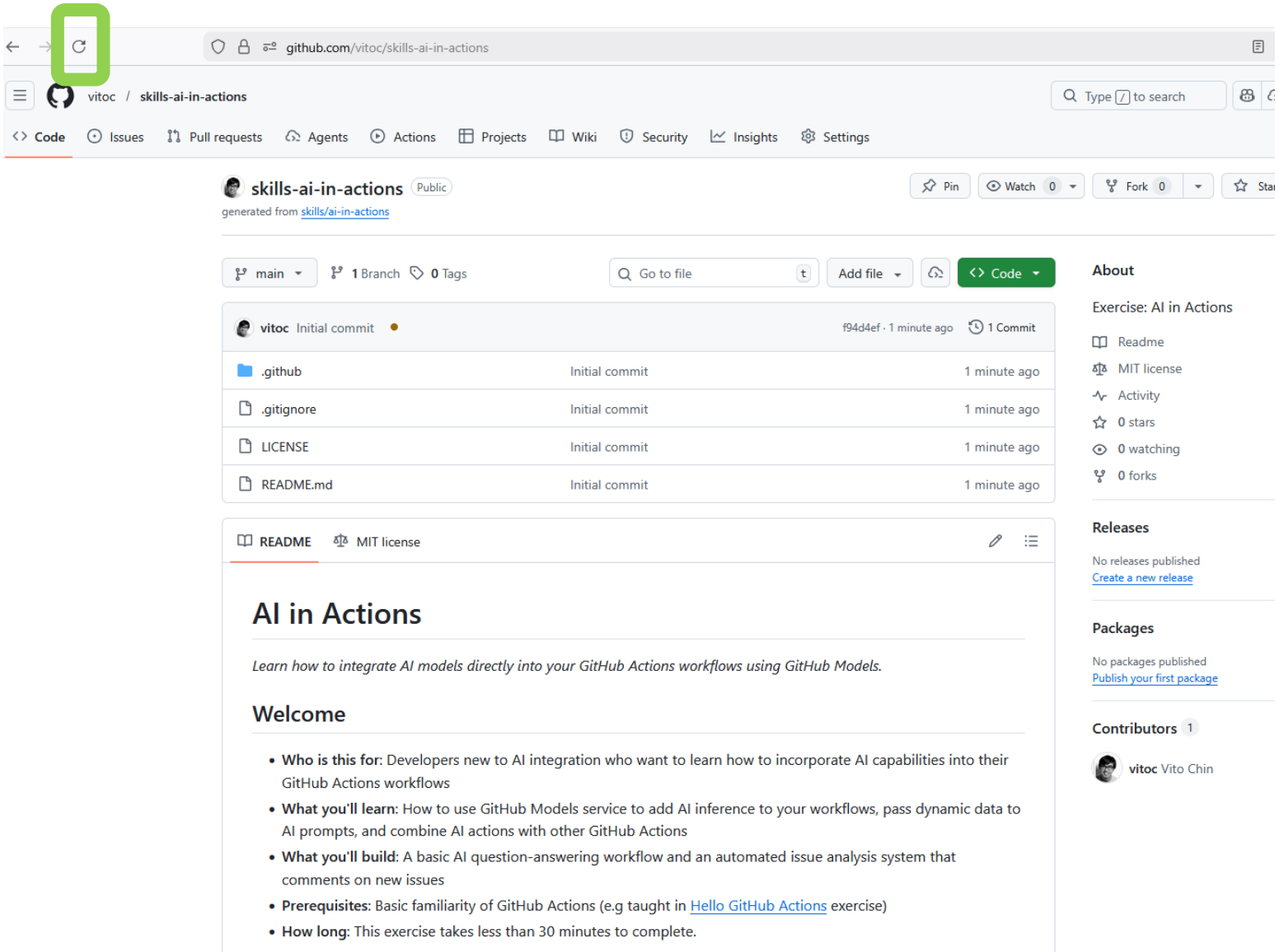
Prompt

Describe what you want Copilot to build

0 / 30000 characters

Create repository

5 – 10초 기다리면


github.com/vitoc/skills-ai-in-actions
vitoc / skills-ai-in-actions
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security
Insights
Settings
skills-ai-in-actions
Public
generated from skills/ai-in-actions
Pin
Watch 0
Fork 0
main
1 Branch
0 Tags
Go to file
Add file
Code
vitoc Initial commit
f94d4ef · 1 minute ago
1 Commit
.github
Initial commit
1 minute ago
.gitignore
Initial commit
1 minute ago
LICENSE
Initial commit
1 minute ago
README.md
Initial commit
1 minute ago
README
MIT license
AI in Actions
Learn how to integrate AI models directly into your GitHub Actions workflows using GitHub Models.
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Who is this for: Developers new to AI integration who want to learn how to incorporate AI capabilities into their GitHub Actions workflows
What you'll learn: How to use GitHub Models service to add AI inference to your workflows, pass dynamic data to AI prompts, and combine AI actions with other GitHub Actions
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How long: This exercise takes less than 30 minutes to complete.
About
Exercise: AI in Actions
Readme
MIT license
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
vitoc Vito Chin

GO TO EXECISE 클릭

skills-ai-in-actions Public

generated from skills/ai-in-actions

Pin Watch 0

main 1 Branch 0 Tags Go to file t Add file Code

github-actions[bot] Start exercise 9117f31 · 1 minute ago 2 Commits

.github	Initial commit	1 minute ago
.gitignore	Initial commit	1 minute ago
LICENSE	Initial commit	1 minute ago
README.md	Start exercise	1 minute ago

README MIT license

AI in Actions

Hey vitoc!

Mona here. I'm done preparing your exercise. Hope you enjoy! 💚

Remember, it's self-paced so feel free to take a break! 🍵

GO TO EXERCISE →

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What is GitHub Models?

- **GitHub 모델** 은 주요 모델 프로바이더들의 AI 모델을 엄선하여 제공하는 서비스입니다.

- 다양한 사용 사례 중 GitHub 모델에는 https://models.github.ai/inference 에서 사용할 수 있는 추론(inference) API가 포함되어 있어, 개발자들이 AI 기능을 GitHub 워크플로우와 애플리케이션에 직접 통합할 수 있습니다.

How GitHub Actions work with GitHub Models

- **빌트인 인증**: GitHub Actions의 내장 GITHUB_TOKEN을 사용하여 GitHub Models 서비스에 대한 호출을 인증할 수 있어, 추가 API Key나 3rd Party와의 복잡한 인증 설정이 필요 없습니다.
- **간단한 권한**: models: read 권한은 GITHUB_TOKEN이 AI 요청을 위해 GitHub Models의 Inference API에 접근할 수 있도록 허용합니다.
- **쉬운 통합**: 공식 actions/ai-inference action은 GitHub Actions에서 GitHub Models를 사용하는 매우 간단한 경로를 제공합니다.

vitoc / skills-ai-in-actions

Type / to search

Code | Issues 1 | Pull requests | Agents | Actions | Projects | Wiki | Security | Insights | Settings

Files

main

Go to file t

- .github
  - steps
  - workflows
    - 0-start-exercise.yml
    - 1-step.yml
    - 2-step.yml
- .gitignore
- LICENSE
- README.md

skills-ai-in-actions / .github / workflows /

Add file

+ Create new file

Upload files

History

vitoc Initial commit

Name	Last commit message	Last commit date
..		
0-start-exercise.yml	Initial commit	5 minutes ago
1-step.yml	Initial commit	5 minutes ago
2-step.yml	Initial commit	5 minutes ago

skills-ai-in-actions / .github / workflows / ask-ai.yml in main

Cancel changes | Commit changes...

Edit | Preview | Spaces | 2 | No wrap

```yaml
name: Ask AI
on:
  workflow_dispatch:

permissions:
  models: read

jobs:
  ask-ai:
    runs-on: ubuntu-latest

    steps:
      - name: AI Inference
        id: ai-response
        uses: actions/ai-inference@v2
        with:
          token: ${{ secrets.GITHUB_TOKEN }}
          prompt: |
            Give me a programming joke.

      - name: Display AI Response
        run: |
          echo "## 🤖 AI Response" >> $GITHUB_STEP_SUMMARY
          echo "" >> $GITHUB_STEP_SUMMARY
          echo "${{ steps.ai-response.outputs.response }}" >> $GITHUB_STEP_SUMMARY
```

Marketplace | Documentation

Search Marketplace for Actions

Featured Actions

Cache ☆ 5.3k
By actions
Cache artifacts like dependencies and build outputs to improve workflow execution time

Setup Node.js environment ☆ 4.7k
By actions
Setup a Node.js environment by adding problem matchers and optionally downloading and adding it to the PATH

Upload a Build Artifact ☆ 4k
By actions
Upload a build artifact that can be used by subsequent workflow steps

Download a Build Artifact ☆ 1.8k
By actions
Download a build artifact that was previously uploaded in the workflow by the upload-artifact action

First interaction ☆ 854

vitoc / skills-ai-in-actions

Type / to search

Code | Issues 1 | Pull requests | Agents | Actions | Projects | Wiki | Security | Insights | Settings

Actions

New workflow

All workflows

Ask AI

Step 1

Step 0 Disabled

Step 2 Disabled

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

Ask AI

ask-ai.yml

Filter workflow runs

Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

0 workflow runs

Event | Status | Branch | Actor

This workflow has a `workflow_dispatch` event trigger.

Run workflow

Use workflow from

Branch: main

Run workflow

This workflow has no runs yet.

github-actions bot 26 minutes ago – with GitHub Actions · edited by github-actions

Edits Contributor Author ···

Step 1 - Passed ✅

Status	Description
✅ - Pass	Ask AI workflow run completed successfully

Theory: Composing AI Workflows

- Actions에서 AI는 세 가지 연속적인 프로세스를 연결하여 지능형 자동화를 생성할 때 가장 큰 가치를 제공합니다:

Workflow 패턴

- **Context 수집**: GitHub Actions를 사용하여 Event Context, 파일 콘텐츠, API 결과 또는 이전 워크플로 단계의 출력을 수집합니다.
- **AI 프로세싱**: 수집된 Context를 actions/ai-inference에 전달하여, AI의 응답을 정의하는 집중된 시스템 프롬프트를 사용하고, 이 응답을 댓글 게시, 파일 업데이트 또는 릴리스 노트 작성과 같은 의미 있는 변경을 생성하는 다른 액션의 입력으로 활용합니다.
- **결과 적용**: AI의 응답을 다른 액션에 입력으로 사용하여 댓글 게시, 파일 업데이트 또는 릴리스 노트 작성과 같은 의미 있는 변화를 만듭니다.

Files

main

Go to file t

- .github
  - steps
  - workflows
    - 0-start-exercise.yml
    - 1-step.yml
    - 2-step.yml
    - ask-ai.yml
  - .gitignore
  - LICENSE
  - README.md

skills-ai-in-actions / .github / workflows / issue-completeness.yml in main

Cancel changes | Commit changes...

Edit | Preview | Spaces | 2 | No wrap

```yaml
name: Issue Completeness

on:
  issues:
    types: [opened]

permissions:
  models: read
  issues: write

jobs:
  analyze:
    name: AI Issue Completeness
    runs-on: ubuntu-latest
    steps:
      - name: Analyze issue with AI
        id: ai-response
        uses: actions/ai-inference@v2
        with:
          token: ${{ secrets.GITHUB_TOKEN }}
          max-tokens: 1000
          system-prompt: |
            You are a GitHub issue assistant. Your task is to analyze newly opened issues for completeness.

            Provide concise, helpful suggestions, ask clarifying questions and identify any missing information that would help resolv

            Always respond with ready-to-use markdown content (no code blocks) that can be posted directly as an issue comment.

            Sign off as AI assistant.
          prompt: |
            New issue was opened by ${{ github.event.issue.user.login }}
            Title: ${{ github.event.issue.title }}
            Body:
            ---
            ${{ github.event.issue.body }}
            ---
      - name: Comment results on the issue
        uses: peter-evans/create-or-update-comment@v4
        with:
          token: ${{ secrets.GITHUB_TOKEN }}
          issue-number: ${{ github.event.issue.number }}
          body: ${{ steps.ai-response.outputs.response }}
```

Marketplace | Documentation

Search Marketplace for Actions

Featured Actions

Cache ☆ 5.3k
By actions
Cache artifacts like dependencies and build outputs to improve workflow execution time

Setup Node.js environment ☆ 4.7k
By actions
Setup a Node.js environment by adding problem matchers and optionally downloading and adding it to the PATH

Upload a Build Artifact ☆ 4k
By actions
Upload a build artifact that can be used by subsequent workflow steps

Setup .NET Core SDK ☆ 1.1k
By actions
Used to build and publish .NET source. Set up a specific version of the .NET and authentication to private NuGet repository

First interaction ☆ 854
By actions
Greet new contributors when they create their first issue or open their first pull request

Featured categories

Code quality | Monitoring
Continuous integration | Project management
Deployment | Testing

Browse all actions on the GitHub Marketplace

Create new issue

Add a title *

Login form throwing 500 errors on mobile

Add a description

Write | Preview

Getting 500 errors when trying to log in on my phone. It works sometimes but not always.

Paste, drop, or click to add files

Write with Copilot

Create more

Cancel

Create

Code | Issues 2 | Pull requests | Agents | Actions | Projects | Wiki | Security | Insights | Settings

Actions

New workflow

All workflows

Ask AI

Issue Completeness

Step 2

Step 0 — Disabled

Step 1 — Disabled

Management

- Caches
- Attestations
- Runners
- Usage metrics
- Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs

Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

5 workflow runs

Event | Status | Branch | Actor

Step 2 Step 2 #1: completed by vitoc		now *Queued*	...
Login form throwing 500 errors on mobile Issue Completeness #1: Issue #2 opened by vitoc		now 10s	...
Step 1 Step 1 #1: completed by vitoc		19 minutes ago 31s	...
Ask AI Ask AI #1: Manually run by vitoc	main	19 minutes ago 10s	...
Initial commit Step 0 #1: Commit f94d4ef pushed by vitoc	main	45 minutes ago 27s	...

github-actions bot 25 minutes ago – with GitHub Actions Contributor Author ···

Step 2 - Passed ✅

Status	Description
✅ - Pass	Issue Completeness workflow completed successfully

github-actions bot 11 minutes ago – with GitHub Actions Contributor Author ···

Review

Congratulations! You've completed this exercise and learned how to integrate AI capabilities into GitHub Actions workflows using GitHub Models.

Here's a recap of your accomplishments:

- Created your first AI-powered GitHub Actions workflow using the `actions/ai-inference` action
- Built an intelligent workflow that automatically responds to new issues with AI-generated feedback
- Integrated multiple actions to create end-to-end automation that combines AI with traditional GitHub Actions

What's next?

- Take another GitHub Skills exercise on GitHub Learn
- Browse the GitHub Marketplace to discover actions that complement `ai-inference` and build more sophisticated AI-powered workflows
- Take a look at other, real world usage examples that are available in the github-samples/models-in-actions repository

github-actions bot 11 minutes ago – with GitHub Actions Contributor Author ···

Congratulations @vitoc! You finished the exercise! 🎉 🎉 🎉

We've updated the repository with a couple changes to highlight your success!

Return to the repository home page to see your progress!